MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.4: Local linearization and differentiability

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Tangent planes and scalar local linearization

Linear approximations and differentiability

Recall that the linear approximation of a function $f : \mathbb{R} \to \mathbb{R}$ of one variable at some point a is the affine function $L(x) = f(a) + f'(a)(x - a)$, and this function gives a local affine approximation of f near a:

$$f(x) \approx f(a) + f'(a)(x - a).$$

Similarly, the partial derivatives of a function can be used to define a local approximation to a function of several variables.

Definition: Linear Approximation

Given a scalar-valued function f, and a point $\mathbf{a}$ in the domain of f where the partial derivatives of f are well-defined,

the [ ] of f at the point $\mathbf{a}$ is

the [ ]

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a})^T(\mathbf{x} - \mathbf{a}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a}) \cdot (\mathbf{x} - \mathbf{a}),$$

where we can expand the dot product that occurs in the equation, so that

$$L(\mathbf{x}) = f(\mathbf{a}) + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{a})(x_i - a_i).$$

In particular, the linear approximation of a scalar-valued function of two variables $f(x, y)$ at a point (a, b) is given by

$$L(x, y) = f(a, b) + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b).$$

The approximation is affine in $\mathbf{x}$; its [ ] is the map $\mathbf{h} \mapsto (\nabla f)(\mathbf{a})^T \mathbf{h}$ on small input changes.

Note

With the base point $\mathbf{a}$ fixed, the gradient $\nabla f(\mathbf{a})$ is a fixed vector. By Linear functions are dot products, its dot product with $\mathbf{h}$ defines a linear function of the input change $\mathbf{h}$:

$$\ell_{\mathbf{a}}(\mathbf{h}) = \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

This is the linear part of the approximation. Taking $\mathbf{x} = \mathbf{a} + \mathbf{h}$, we can write

$$L(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \ell_{\mathbf{a}}(\mathbf{h}).$$

Now suppose a scalar-valued function f of a single variable is differentiable at a point a, i.e., so that the limit

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

exists. For $x \neq a$, subtracting $f'(a)$ from the difference quotient and using the common denominator $x - a$ gives

$$\frac{f(x)-f(a)}{x-a} - f'(a) = \frac{f(x)-f(a)}{x-a} - \frac{f'(a)(x-a)}{x-a} = \frac{f(x)-f(a)-f'(a)(x-a)}{x-a}.$$

Therefore,

$$\lim_{x\to a} \frac{f(x)-f(a)-f'(a)(x-a)}{x-a} = 0,$$

or, recognizing the numerator as $f(x) - L(x)$,

$$\lim_{x\to a} \frac{f(x)-L(x)}{x-a} = 0,$$

where $L(x) = f(a) + f'(a)(x-a)$ is the linear approximation of f at a. A function of several variables is differentiable when the same property holds, with the linear approximation given by the partial derivatives of the function.

Definition: Differentiability

A scalar-valued function f is $\boxed{}$ at a point $\mathbf{a}$ if the gradient of f is well-defined at $\mathbf{a}$, and if

$$\lim_{\mathbf{x}\to\mathbf{a}} \frac{f(\mathbf{x})-L(\mathbf{x})}{\|\mathbf{x}-\mathbf{a}\|} = 0,$$

where

$$L(\mathbf{x}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a})^T(\mathbf{x}-\mathbf{a}) = f(\mathbf{a}) + (\nabla f)(\mathbf{a}) \cdot (\mathbf{x}-\mathbf{a})$$

is the linear approximation of f at $\mathbf{a}$. A vector-valued function is differentiable at a point precisely when each of its component functions is differentiable at that point.

Note

The equation

$$\lim_{\mathbf{x}\to\mathbf{a}} \frac{f(\mathbf{x}) - L(\mathbf{x})}{\|\mathbf{x} - \mathbf{a}\|} = 0$$

can only hold if L is a *good* linear approximation of f near $\mathbf{a}$, so that the magnitude of the numerator is much smaller than the magnitude of the denominator when $\mathbf{x}$ is near $\mathbf{a}$, i.e., when $\|\mathbf{x} - \mathbf{a}\|$ is small.

The following theorem is the main way we will determine if a function is differentiable.

Theorem: Continuity of First Partials Implies Differentiability

Let f be a scalar-valued function, and suppose $\mathbf{a}$ is an interior point (recall Interior, boundary, and closed sets) of the domains of f and its partial derivatives. Moreover, suppose that all of the partial derivatives $f_{x_1}, \ldots, f_{x_n}$ are continuous at $\mathbf{a}$. Then f is differentiable at $\mathbf{a}$.

Why is this true?

Here is the intuition in two variables. To move from (a, b) to a nearby point $(a + h, b + k)$, change one coordinate at a time:

$$(a, b) \longrightarrow (a + h, b) \longrightarrow (a + h, b + k).$$

On each step, only one variable changes, so we can use a one-variable linear approximation. Continuity of the partial derivatives means that the slopes along these small steps stay close to their values at (a, b). Thus

$$\begin{aligned} f(a + h, b) - f(a, b) &\approx f_x(a, b)h, \\ f(a + h, b + k) - f(a + h, b) &\approx f_y(a, b)k. \end{aligned}$$

Adding the two changes cancels the value at the intermediate point and gives

$$f(a + h, b + k) - f(a, b) \approx f_x(a, b)h + f_y(a, b)k.$$

Adding $f(a,b)$ to both sides gives an approximation for the value at the new point:

$$f(a+h,b+k) \approx f(a,b) + f_x(a,b)h + f_y(a,b)k.$$

This explains both the sum and its coefficients: each partial derivative measures the change per unit step in one coordinate. Continuity is what lets us use the slopes at the base point throughout the nearby steps; each small error in a slope is multiplied by the corresponding small coordinate change.

In n variables, the same idea uses n coordinate steps. Adding their contributions gives the linear approximation

$$\begin{aligned} L(\mathbf{a}+\mathbf{h}) &= f(\mathbf{a}) + \sum_{i=1}^{n} f_{x_i}(\mathbf{a})h_i \\ &= f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h}. \end{aligned}$$

Activity: Linear approximation in three variables (U3-LO4)

Consider the function

$$f(x,y,z) = x^2y + yz^2.$$

Find the linear approximation of f at the point $(1,2,0)$.

Activity: Linear approximation in three variables (U3-LO4)

Consider the function

$$f(x, y, z) = x^2y + yz^2.$$

Use the linear approximation you have found to approximate the value $f(1.04, 1.97, 0.05)$.

Activity: Predicting a change in fitting error (U3-LO4)

For the prediction loss in From predictions to a scalar error function (U3-LO1, U2-LO1), How model weights change the loss (U3-LO3) gives

$$\mathscr{L}(\mathbf{w}_0) = 1, \qquad \nabla\mathscr{L}(\mathbf{w}_0) = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \qquad \mathbf{w}_0 = (1, 1).$$

Suppose we increase both weights by $1/10$, so $\mathbf{h} = (1/10, 1/10)$.

1. Find the affine approximation $\widehat{\mathscr{L}}(\mathbf{w})$ near $\mathbf{w}_0$. Use it to predict the change in loss and the new loss at $\mathbf{w}_0 + \mathbf{h}$.
2. Compute the new predictions $X(\mathbf{w}_0 + \mathbf{h})$, residuals, and exact loss. Compare the exact loss with the affine prediction.
3. For an arbitrary change $\mathbf{h} = (h_1, h_2)$, expand $\mathscr{L}(\mathbf{w}_0 + \mathbf{h})$. Show that the approximation error is $\|X\mathbf{h}\|^2$. What happens to this error if the change is halved?



Tangent planes

Activity: A plane containing two lines (U2-LO1)

Find an equation for the plane containing the lines L_1 and L_2, where L_1 is the line parameterized by the function

$$\mathbf{r}_1(t) = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + t \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

and where L_2 is the line parameterized by the function

$$\mathbf{r}_2(s) = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} + s \begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix}.$$

Definition: Tangent Plane

Let S be a surface, and $\mathbf{x}_0$ a point on S. A [ ] to S at $\mathbf{x}_0$ is a plane that contains the tangent lines of any differentiable curve contained in S as it passes through $\mathbf{x}_0$.

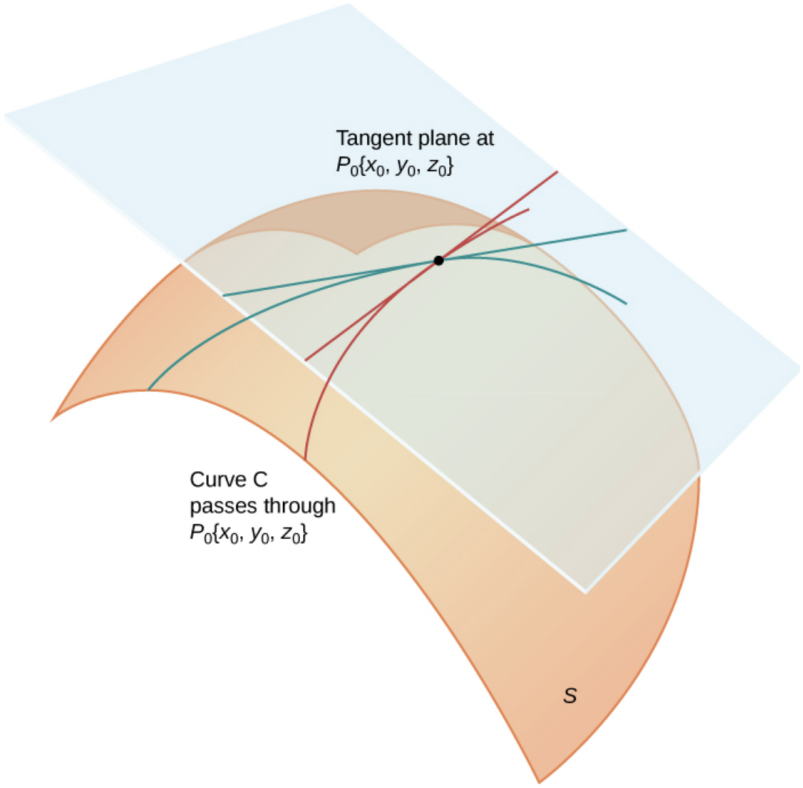


Figure. A tangent plane touching a surface along tangent lines through a point.

Theorem: Equation of the Tangent Plane

Suppose S is the graph of a function $f(x, y)$ which is differentiable at a point (x_0, y_0). Then the tangent plane to S exists, is unique, and is described by the equation

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Note that this plane is the graph of the linear approximation L to f at (x_0, y_0).

Why is this true?

We verify that the plane described by the equation above is the only possible tangent plane. Recall from Geometric Interpretation that the intersection of S with the plane $y = y_0$ is the curve $z = f(x, y_0)$, whose slope at $x = x_0$ is $f_x(x_0, y_0)$. Its tangent line therefore has equations

$$y = y_0, \qquad z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0).$$

Set $t = x - x_0$, so that $x = x_0 + t$. The line can then be parameterized as

$$\mathbf{r}_1(t) = \begin{bmatrix} x_0 \\ y_0 \\ f(x_0, y_0) \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ f_x(x_0, y_0) \end{bmatrix}.$$

The vector multiplying t is a direction vector: a change of t in x leaves y fixed and changes z by $f_x(x_0, y_0)t$. This gives the direction vector $(1, 0, f_x(x_0, y_0))$.
Similarly, the intersection with $x = x_0$ is the curve $z = f(x_0, y)$, whose tangent line has equations

$$x = x_0, \qquad z = f(x_0, y_0) + f_y(x_0, y_0)(y - y_0).$$

Taking $s = y - y_0$ gives the parameterization

$$\mathbf{r}_2(s) = \begin{bmatrix} x_0 \\ y_0 \\ f(x_0, y_0) \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ f_y(x_0, y_0) \end{bmatrix}.$$

Here x stays fixed while y changes by s and z changes by $f_y(x_0, y_0)s$, so the direction vector is $(0, 1, f_y(x_0, y_0))$.
Following the method of A plane containing two lines (U2-LO1), write a normal vector as $\mathbf{n} = (a, b, c)$ and require it to be perpendicular to both direction vectors. Taking the two dot products gives the homogeneous system

$$\begin{aligned} a + f_x(x_0, y_0)c &= 0, \\ b + f_y(x_0, y_0)c &= 0. \end{aligned}$$

With the variables in the order a, b, c, its augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & f_x(x_0, y_0) & 0 \\ 0 & 1 & f_y(x_0, y_0) & 0 \end{array}\right].$$

This matrix is already in row echelon form (in fact, reduced row echelon form), so no row operations are needed. The first two columns are

pivot columns: a and b are basic variables, while c is free. Reading the equations gives

$$a = -f_x(x_0, y_0)c, \qquad b = -f_y(x_0, y_0)c.$$

Setting the free variable $c = 1$ gives the basic solution

$$\mathbf{n} = \begin{bmatrix} -f_x(x_0, y_0) \\ -f_y(x_0, y_0) \\ 1 \end{bmatrix}.$$

This vector is nonzero and perpendicular to both tangent-line directions, so it is a normal vector to the plane containing the two lines.
Thus the tangent plane to S at $(x_0, y_0, f(x_0, y_0))$ must be described by the equation

$$(-f_x(x_0, y_0), -f_y(x_0, y_0), 1) \cdot (x - x_0, y - y_0, z - f(x_0, y_0)) = 0,$$

an equation we may rearrange to read

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Activity: A tangent plane to a surface (U3-LO4)

Find an equation of the tangent plane to the surface

$$z = e^x \cos y$$

at $(0, \frac{\pi}{2}, 0)$.



An important property of the gradient of a function at a point is that it is a vector that is perpendicular to the level curves of the function. We can now use tangent planes to explain this property.

Theorem

Let $f(x, y)$ be a function with continuous partial derivatives. If (a, b) is a point where $\nabla f(a, b) \neq \mathbf{0}$, then $\nabla f(a, b)$ is

 of f passing through the point (a, b).

Why is this true?

By Continuity of First Partials Implies Differentiability, the continuous partial derivatives make f differentiable at (a, b). Let $c = f(a, b)$. By Equation of the Tangent Plane, the tangent plane to the graph of f at (a, b, c) has equation

$$z = c + \frac{\partial f}{\partial x}(a, b)(x - a) + \frac{\partial f}{\partial y}(a, b)(y - b).$$

Along the level curve $f(x, y) = c$, the height is constant. Lifting this curve onto the graph of f therefore places it entirely in the horizontal plane $z = c$. If $\mathbf{v} = (v_1, v_2)$ is a nonzero tangent direction to the level curve at (a, b), then the lifted curve has tangent direction $(v_1, v_2, 0)$: there is no vertical change.

The tangent line to the lifted curve passes through (a, b, c) and has direction vector $(v_1, v_2, 0)$, so its vector parameterization is

$$\mathbf{r}(t) = \begin{bmatrix} a \\ b \\ c \end{bmatrix} + t \begin{bmatrix} v_1 \\ v_2 \\ 0 \end{bmatrix}, \qquad t \in \mathbb{R}.$$

This line lies in the tangent plane, so its coordinates $x = a + tv_1$, $y = b + tv_2$, and $z = c$ satisfy the plane equation. Substituting these expressions and subtracting c from both sides gives

$$0 = t\left(\frac{\partial f}{\partial x}(a, b)v_1 + \frac{\partial f}{\partial y}(a, b)v_2\right).$$

For $t \neq 0$, dividing by t yields

$$\nabla f(a, b) \cdot \mathbf{v} = \frac{\partial f}{\partial x}(a, b)v_1 + \frac{\partial f}{\partial y}(a, b)v_2 = 0.$$

A zero dot product means that the gradient is perpendicular to the tangent direction of the level curve.

See the figure for a graphical example of the theorem.

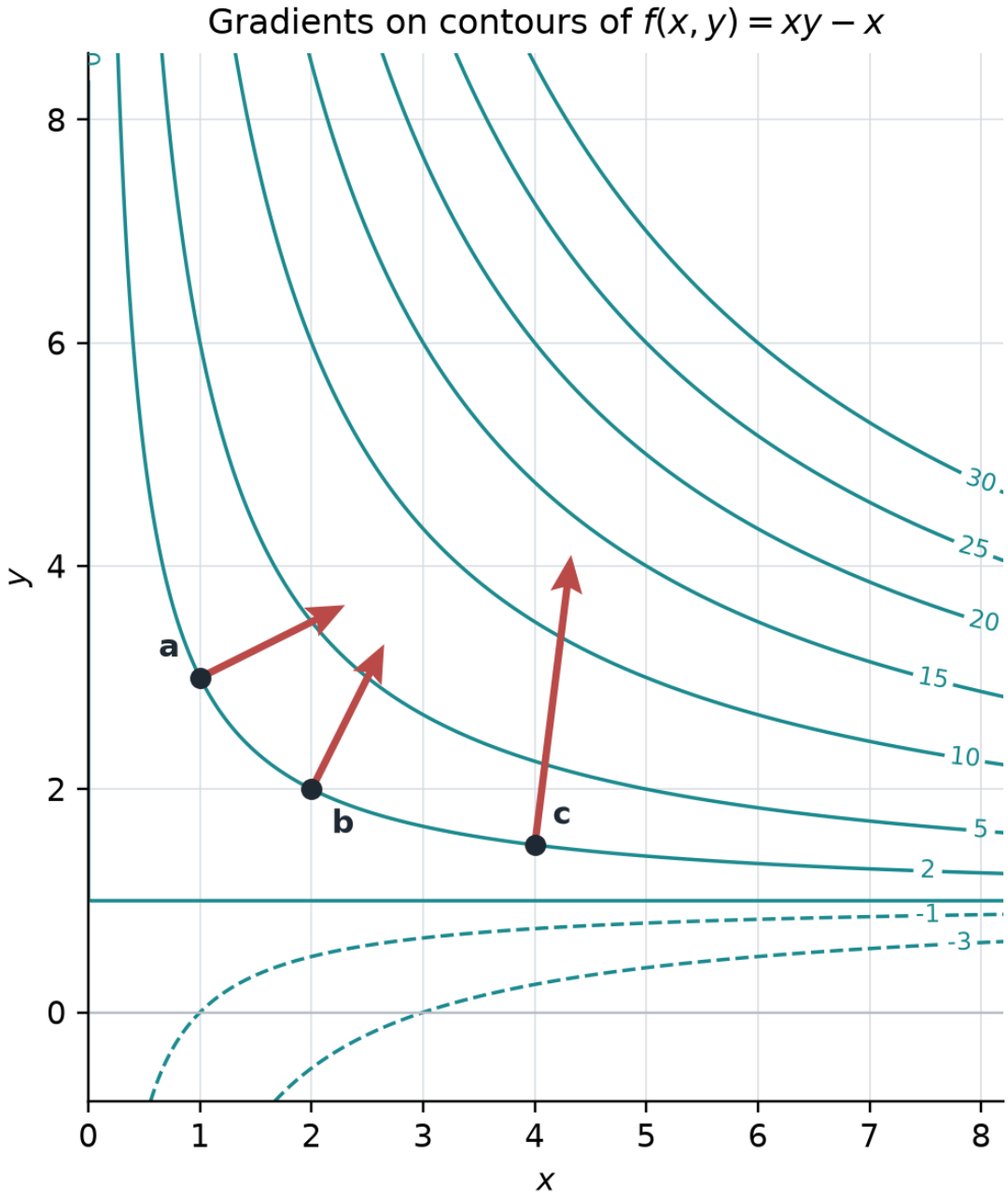


Figure. A contour plot of the function $f(x, y) = xy - x$, with the gradient vectors at three points $\mathbf{a} = (1, 3)$, $\mathbf{b} = (2, 2)$, and $\mathbf{c} = (4, 3/2)$. Adapted from: Stanford's MATH 51 textbook

Vector-valued local linearization

Linear maps reappear locally

A nonlinear map is usually not linear globally. Near one input, however, differentiability replaces small input changes by a linear map:

$$F(\mathbf{a}+\mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

$\mathbf{x} \mapsto F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$ is affine in $\mathbf{x}$. $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ is the linear map on input changes.

Definition: Linear Approximation of Vector-Valued Functions

Let $F : \mathbb{R}^n \to \mathbb{R}^m$ be a vector-valued function. Provided that the partial derivatives of the components of F all exist at $\mathbf{a}$, so that the Jacobian matrix J_F is well-defined,

the [ ] of F at $\mathbf{a}$ is the vector-valued affine function

$$L(\mathbf{x}) = F(\mathbf{a}) + J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$$

where $J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$ is the Jacobian matrix of F evaluated at $\mathbf{a}$, and then applied to the vector $\mathbf{x}-\mathbf{a}$. The approximation is affine in $\mathbf{x}$; its linear part is the map $\mathbf{h} \mapsto J_F(\mathbf{a})\mathbf{h}$ on small input changes.

Note: Componentwise linear approximation

Recall from Jacobian matrices that $J_F(\mathbf{a})$ is the $m \times n$ matrix whose (i,j)-entry is $\frac{\partial F_i}{\partial x_j}(\mathbf{a})$. Its ith row is the transpose of the gradient of the component function F_i:

$$\left[\frac{\partial F_i}{\partial x_1}(\mathbf{a}) \quad \cdots \quad \frac{\partial F_i}{\partial x_n}(\mathbf{a})\right] = (\nabla F_i)(\mathbf{a})^T.$$

Multiplying this row by $\mathbf{x}-\mathbf{a}$ gives the ith component of $J_F(\mathbf{a})(\mathbf{x}-\mathbf{a})$, so

$$L_i(\mathbf{x}) = F_i(\mathbf{a}) + (\nabla F_i)(\mathbf{a})^T(\mathbf{x}-\mathbf{a}) = F_i(\mathbf{a}) + (\nabla F_i)(\mathbf{a}) \cdot (\mathbf{x}-\mathbf{a}).$$

Thus the components of L are precisely the scalar-valued linear approximations of the components of F.

Activity: Approximating a two-component map (U3-LO5)

Let $F(x, y) = (x^2 y, xe^y)$. Find the linear approximation of F at the point $(1, 0)$, and use it to approximate $F(1.1, -0.2)$.

Activity: Local square-grid visualization (U3-LO5)

Return to

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

The Jacobian matrix is

Activity: Local square-grid visualization (U3-LO5) (continued)

$$J_F(x, y) = \begin{bmatrix} 1 & 2y \\ 0 & 1 \end{bmatrix}.$$

Let

$$\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}.$$

1. Compute $J_F(\mathbf{a})$.
2. Let $\mathbf{h} = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$. Compute $F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a})$.
3. Compute $J_F(\mathbf{a})\mathbf{h}$.
4. Compute the error

$$F(\mathbf{a} + \mathbf{h}) - F(\mathbf{a}) - J_F(\mathbf{a})\mathbf{h}.$$

5. If $0 \le h_2 \le r$, how large can the first coordinate of the error be?
6. What happens to the local square-grid picture as r gets smaller?

Note

The local square-grid visualization does not produce one matrix for the whole nonlinear map. It shows one matrix approximation near one base point.

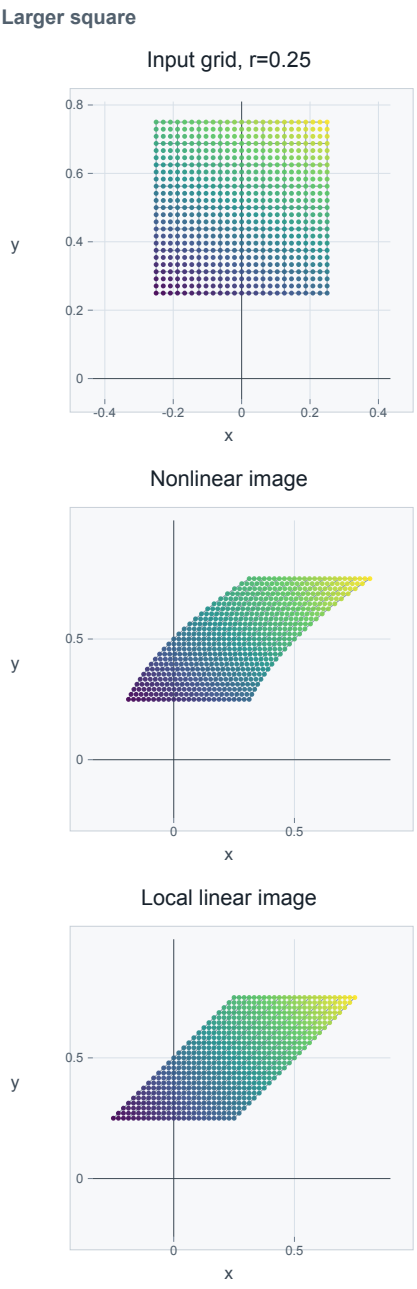


Figure. Local square-grid visualization near $\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ for the larger input square $r = 0.25$. The three panels show the input grid, the nonlinear image, and the local linear image on ticked axes.

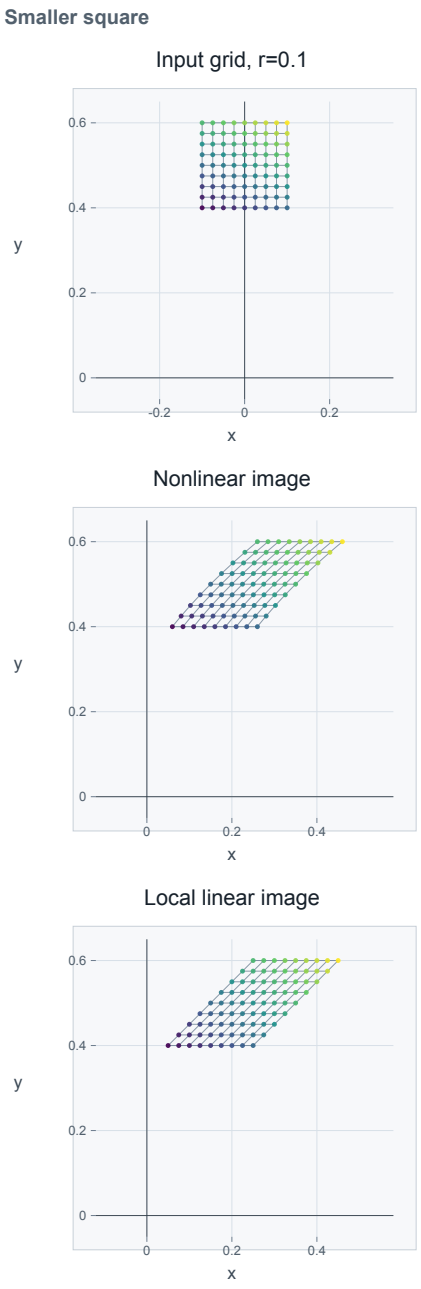


Figure. Local square-grid visualization near $\mathbf{a} = \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ for the smaller input square $r = 0.1$. The nonlinear image is closer to the local linear image than in the larger-square visualization.

Activity: Reading a local linear prediction in SymPy (U3-LO5)

Use SymPy to revisit the nonlinear shear $F(x, y) = (x + y^2, y)$ from Local square-grid visualization (U3-LO5). We evaluate its Jacobian at $\mathbf{a} = (0, 1/2)$, then use that matrix to predict the output after a

Activity: Reading a local linear prediction in SymPy (U3-LO5) (continued)

small input change. The command `sp.Rational(p, q)` represents the fraction p/q exactly. As in Using a SymPy null-space vector (U2-LO3, U2-LO4, U2-LO6), `*` between SymPy matrices means matrix multiplication.

```
import sympy as sp

x, y = sp.symbols("x y")
F = sp.Matrix([x + y**2, y])
a = sp.Matrix([0, sp.Rational(1, 2)])
h = sp.Matrix([sp.Rational(1, 10), -sp.Rational(1, 10)])

J = F.jacobian([x, y])
Ja = J.subs({x: a[0], y: a[1]})
Fa = F.subs({x: a[0], y: a[1]})
prediction = Fa + Ja*h
actual = F.subs({x: a[0] + h[0], y: a[1] + h[1]})

print(Ja)
print(Ja*h)
print(prediction)
print(actual)
print(actual - prediction)
```

Output:

```
Matrix([[1, 1], [0, 1]])
Matrix([[0], [-1/10]])
Matrix([[1/4], [2/5]])
Matrix([[13/50], [2/5]])
Matrix([[1/100], [0]])
```

1. What are the shapes of `Ja`, `h`, and `Ja*h`? Identify the shear matrix from Horizontal shear (U1-LO4). Why does `Ja*h` represent a predicted change rather than the predicted output?

Activity: Reading a local linear prediction in SymPy (U3-LO5) (continued)

2. Write the matrix calculation represented by `Fa + Ja*h`. Which output coordinate is predicted exactly for this input change, and what is the error in the other coordinate?

3. If `h` is replaced by `h/2` before computing `prediction` and `actual`, predict the new error. Explain the change using the error formula in Local square-grid visualization (U3-LO5).

Activity: Interpreting a Jacobian's null space in SymPy (U3-LO5, U2-LO3)

In Using a SymPy null-space vector (U2-LO3, U2-LO4, U2-LO6), a null-space vector described an input change that a linear map forgets. Apply the same tools to the local linear map of $F(x, y) = (x + y, x^2 + y^2)$ at $\mathbf{a} = (1, 1)$. Recall that `.rref()` returns the reduced matrix and zero-based pivot-column indices, while `.nullspace()` returns a list of basis column matrices.

```
import sympy as sp

x, y = sp.symbols("x y")
F = sp.Matrix([x + y, x**2 + y**2])
J = F.jacobian([x, y])
Ja = J.subs({x: 1, y: 1})

R, pivots = Ja.rref()
basis = Ja.nullspace()
v = basis[0]

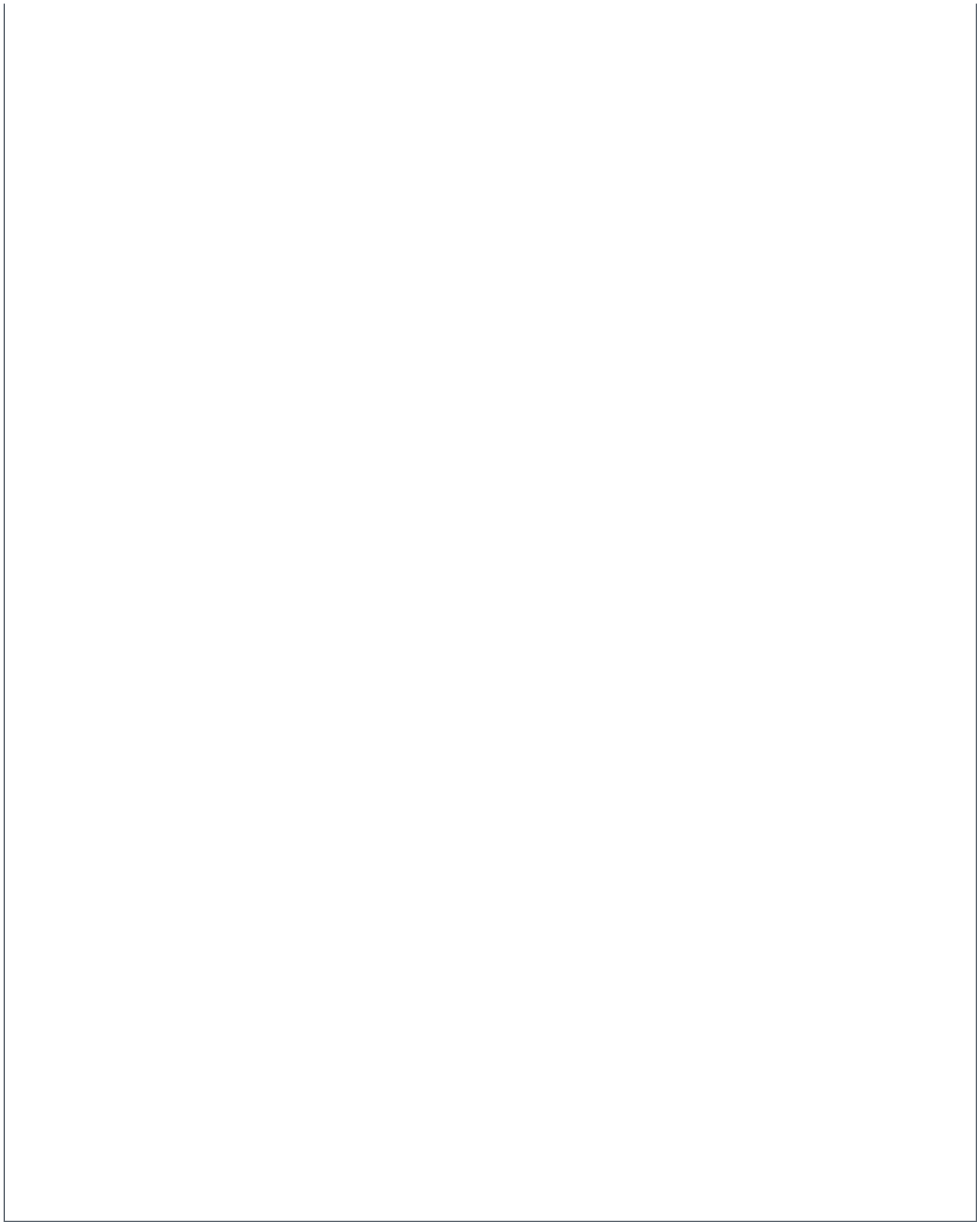
print(Ja)
print(R, pivots)
print(len(basis), v)
print(Ja*v)
```

Output:

Activity: Interpreting a Jacobian's null space in SymPy (U3-LO5, U2-LO3) (continued)

```
Matrix([[1, 1], [2, 2]])
Matrix([[1, 1], [0, 0]]) (0,)
1 Matrix([[-1], [1]])
Matrix([[0], [0]])
```

1. What are the rank and nullity of `Ja`? Using its original pivot column, describe the possible first-order output changes $J_F(\mathbf{a})\mathbf{h}$.
2. Let $\mathbf{v} = (-1, 1)$. What does the local linear approximation predict for $F(\mathbf{a} + t\mathbf{v})$ when t is small?
3. Substitute $\mathbf{a} + t\mathbf{v} = (1 - t, 1 + t)$ directly into F. Does the zero product `Ja*v` imply that the nonlinear output stays exactly constant? Find the actual change when $t = 1/10$.
4. Predict `J.subs({x: 1, y: 0}).nullspace()`. What does this tell you about how the local linear map depends on the base point?



The approximation in this unit is local. Near one input $\mathbf{a}$, a differentiable function behaves like

$$F(\mathbf{a}+\mathbf{h}) \approx F(\mathbf{a}) + J_F(\mathbf{a})\mathbf{h}.$$

Unit 4 studies a different kind of approximation:

$$\min_x \|Ax - b\|.$$

That is a global best approximation by a vector in a subspace. Unit 3 asks what a nonlinear function looks like near one point; Unit 4 asks which allowable vector is closest overall.